

Amin Mavaddat

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Profile

Data Scientist with experience in designing and implementing data-driven algorithms, machine learning models, and advanced analytics pipelines for energy, healthcare, and financial industries. Skilled in Python, SQL, and cloud-based platforms for large-scale data processing. Background in statistical analysis, feature engineering, and model validation, with practical experience in inspection, predictive analytics, and real-time sensor/vision data processing.

Professional Experience

AI Consultant 2025/11 – 2026/08 | Vienna-Remote
Predictores.ai- contract
- Implementing a multi-agent (supervisor architecture), self-evolving RAG system on vector databases to reduce token usage.
- Model output evaluation on retrieving sensetive information generation and active filtration.
- Token-aware fine-tuning of foundation models to hinder catastrophic forgetting in LLM models.

Research Assistant 2023/11 – 2025/09 | Lappeenranta, Finland
LUT University, HORIZON Project-SHEREC
- Developed supervised, unsupervised, and deep learning models for anomaly detection and fault classification in industrial time-series and multimodal datasets.
- Built end-to-end VAE-latent Diffusion models to 3D reconstruction and denoising.
- Developed and fine-tuned LLM- and VLLM-based systems (LLaVA-8B, embedding models) for retrieval, ranking, grounding, and structured information extraction from unstructured and multimodal data.
- Analyzed high-resolution RGB data using Vision Transformer (ViT) and Convolutional Neural Network (CNN) architectures for image segmentation and classification.

Research Assistant in Computer vision 2022/02 – 2023/08 | Tonsberg, Norway
University of South Eastern Norway
- Data collection and dataset generation from RGB images and sensor fusion for computer vision (CV) model training.
- Object detection and classification model training and evaluation using deep learning–based computer vision pipelines.
- Lifecycle management and monitoring of trained models during inference and deployment.

Computer vision engineer 2018/01 – 2020/12 | Dubai-Remote
Medicolab
- Developed real-time industrial computer vision systems for weld and cut-line inspection, including surface-defect detection using CNNs and classical feature pipelines.
- Conducted multimodal analysis combining RGB imaging, vibration signals, and strain sensors to model structural behavior and detect anomalies.
- Built automated imaging workflows for material characterization and quality assurance, optimizing preprocessing and camera-sensor calibration.

- Analyzed complex surface patterns using combined sensor–image representations to support predictive maintenance and materials evaluation.

Python Developer 2015/02 – 2017/11 | Tabriz
Ayandeh Bank
- •Created backend solutions for banking data automation.
- Built Python- and SQL-based ETL workflows to automate financial reporting and compliance

Education

M.Sc. in Micro-Nano system technology 2021 – 2023 | Vestfold, Norway
University of South-Eastern Norway
Master Thesis:
Artificial Intelligence application in agriculture: Digital Tree

B.Sc. in Applied Physics 2009 – 2012 | Tabriz,Iran

Publications

A Hybrid Deep Learning-Based Approach in Agricultural Sustainability
2025 3rd Cognitive Models and Artificial Intelligence Conference (AICCONF)

Patch-Wise Regressor-Guided Preprocessing for Multiple Instance Learning in RGB Image Classification
IEEE CAI 2026 Conference on Artificial Intelligence 2026

Submitted PUBLICATION (First author or Equally ocntributed as first author):

2. Fingerprint-Conditioned Diffusion for Zero-Shot Cross-Chip MEMS IMU Calibration
ICASSP 2027

1. Safe Ship Chunk Dismantling After Hazardous Material Removal Using Reinforcement Learning and Convex Hull Methods

Scientific Reports

3. Lightweight and Data-Efficient Image Segmentation Architecture for Limited Annotated Datasets
Neural Computing and Applications

4. A new method to control devices using vision-based hand gesture recognition
Pattern recognition

5. Stress Identification in Excavator Operators Using Deep Learning for Enhanced Human– Machine Interaction in High-Risk Environments
Neural Computing and Applications

Languages

English — Fluent

Farsi — Native/Bilingual

Norwegian — Conversational

Soumi — Basic
Basic

Turkish — Fluent

References

Available upon request.